Q7(a)
Speed of light $= 3.0 \times 10^{8}$ m s$^{-1}$.

Time required for EM waves to travel to Earth $= \frac{s}{v} = \frac{200 \times 10^{6} \times 10^{3}}{3.0 \times 10^{8}} = 666.67$ s

$= 11.1$ mins .

Q7(b)(i)
The reduced gravity of 0.38g of Earth makes it easier to lift off and fly
but the low density of air makes it harder to fly as the rotors have less air to push against.

Q7(b)(ii)
By Newton's 2nd law,
force of rotor blades on air particles = rate of change of momentum of the air

By considering the mass of air passing through the blades as passing through a cylinder, we can write $m = \rho V = \rho A x$, where $x$ the height of the cylinder of air.

$$F = (v - 0)\frac{dm}{dt} = (v - 0)\frac{d(\rho A x)}{dt}$$

Since $\rho$ and $A$ are constants,

$$F = v\rho A\frac{dx}{dt} = \rho A v^{2}$$

This force $F$ is in the downwards direction.
By Newton's 3rd law, the lift force
$F_L = \rho A v^{2}$ in the upwards direction

Q7(b)(iii)
To hover, $F_L = weight$

$$F_L = m(0.38g) = 1.8\,(0.38)(9.81)$$

$$\rho A v^{2} = 1.8\,(0.38)(9.81)$$

$$v^{2} = \frac{1.8(0.38)(9.81)}{\rho A}$$

$$= \frac{1.8(0.38)(9.81)}{(0.020)\left(\pi\left(\frac{1.2}{2}\right)^{2}\right)}$$

$= 296.64$

$$\boldsymbol{v = 17.2\ \mathrm{m\ s^{-1}}}$$

Q7(b)(iv)

Since $f \propto v$

$$f = kv = k\sqrt{\frac{F_L}{\rho A}}$$

On Earth $F_{LEarth} = mg$,

On Mars $F_{LMars} = m(0.38g)$

On Earth $\rho_{Earth} = 1.225 \text{ kg m}^{-3}$,

On Mars $\rho_{Mars} = 0.020 \text{ kg m}^{-3}$

Hence

$$\frac{f_{Mars}}{f_{Earth}} = \frac{k\sqrt{\frac{F_{LMars}}{\rho_{Mars}A}}}{k\sqrt{\frac{F_{LEarth}}{\rho_{Earth}A}}} = \sqrt{\frac{F_{LMars}}{F_{LEarth}} \times \frac{\rho_{Earth}}{\rho_{Mars}}}$$

$$= \sqrt{\frac{0.38}{1} \times \frac{1.225}{0.020}} = 4.824 \approx \mathbf{4.82}$$

Q7(c)(i)

**200 W** from Fig. 7.2

Q7(c)(ii)

Area under the graph between 12 s and 19 s = height

This area is slightly more than that of a triangle of base 7s and velocity 1 s.

Height > 0.5(7)(1).

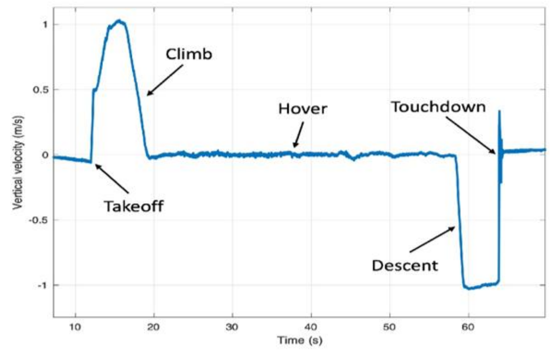
The estimated hovering height is 4.0 m

Q7(c)(iii)

When the motor power dipped below 200 W, the helicopter started to accelerate downwards.

When it reached the desired downward velocity, the rotors spun again so that it can descend at a constant speed.

Q7(c)(iv)

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